Fiscal Year:	FY 2006	**Task Last Updated:**	FY 08/30/2006
PI Name:	Wray, Gregory Ph.D.		
Project Title:	Evolution Of Endo16 Promoter Function In Camarodont Sea Urchins		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Evolutionary biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Invertebrate		
Space Biology Cross-Element Discipline:	(1) Developmental Biology		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Space Biology 01-OBPR-06
Start Date:	11/01/2002	**End Date:**	04/30/2006
No. of Post Docs:	2	**No. of PhD Degrees:**	2
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):			
Grant/Contract No.:	NAG2-1583		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The aim of this project is to understand the molecular bases for evolutionary changes in transcriptional regulation. We are focusing on one of the best-characterized cis-regulatory regions of any eukaryotic gene, the Endo16 locus of camarodont sea urchins. The project has three basic aims: (1) to identify the genetic and molecular changes that have produced functional compensation in promoter function among species, both on a small scale and a more extensive scale; (2) to identify the genetic and molecular changes that have produced a functional difference in transcription profiles among species; and (3) to identify evolutionary changes in promoter function at a biochemical level by testing known and artificial variants of transcription factor binding sites.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Understanding how gene regulatory systems operate and evolve is an essential part of several important Earth benefits. First, it will be fundamental to our understadning of the history of life on Earth, since it provides insights into how genetic systems evolve and what constraints they operate under. Second, it will provide insights into our understanding of the general properties of life: all life requires control systems to extract information from the genetic repository and deploy it in response to the environment.
Task Progress:	The overall goal of this project was to understand the molecular bases for evolutionary changes in transcriptional regulation. We focused on one of the best-characterized cis-regulatory regions of any eukaryotic gene, the Endo16 locus of camarodont sea urchins. The project had three aims: (1) to identify the genetic and molecular changes that have produced functional compensation in promoter function among species, both on a small scale and a more extensive scale; (2) to identify the genetic and molecular changes that have produced a functional difference in transcription profiles among species; and (3) to identify evolutionary changes in promoter function at a biochemical level by testing known and artificial variants of transcription factor binding sites. Major findings Analyses of genetic polymorphism and divergence: Measurements of genetic variation within species and divergence between species were carried out, so as to provide a framework for the experimental goals of this project. In particular, we identified more than 100 single nucleotide polymorphisms (SNPs) and insertion/deletion polymorphisms (indels) within the Endo16 promoter region and assigned these to putative functional classes for later experimental tests. We also carried out quantitative analyses that identified a signature of negative selection to maintain functional binding sites within the proximal but not distal promoter modules, demonstrated a parallel trend in interspecific divergence in promoter sequences, and found that promoter sequences in species more than 40 million years diverged are alignable only over a small fraction of their total length. The polymorphism study has been published (Balhoff and Wray 2005) and the interspecific divergence study is currently in preparation (Balhoff, Fédrigo, and Wray, in prep.). Assays of promoter function in vivo We carried out transient transfection assays to test the function of the Endo16 promoter from L. variegatus in S. purpuratus embryos in a fully complementary experimental design. We showed that despite the high degree of sequence divergence between these two species (see above), the promoter from each functions nearly perfectly when tested in embryos of the other species. We also demonstrated that compensatory mutations in cis (within the Endo16 promoter) and in trans (elsewhere in the genome) together maintain a nearly conserved transcription profile. These results provide one of the most dramatic and clearly documented cases of functional compensation in promoter sequences yet identified, and suggest that mechanisms transcriptional regulation can evolve faster than the transcription profiles they generate. This work was published as Romano and Wray (2003), and has already been featured in at least two books that consider the evolution of transcriptional regulation (Carroll, Grenier and Weatherbee, 2005, From DNA to Diversity; Wagner, 2005, Robustness and Evolvability in Living Systems). Assays of protein:DNA interaction: Based on the sequence comparisons mentioned above, we identified 12 segregating mutations in transcription factor binding sites of the Endo16 promoter within S. purpuratus and 11 additional fixed mutations that distinguish this region of the genome in S. purpuratus and S. droebachiensis (its closest living relative). We measured the allelic difference in the affinity and specificity of protein:DNA binding in vitro for all of these mutations using electromobility shift assays. The most surprising finding from these analyses is that most of the derived alleles and fixed differences result in an increase in binding affinity. Some are relatively small affinity differences (<2-fold) but a few are larger (up to 20-fold). The representation of affinity gains over losses is highly significant statistically, and suggests that positive selection plays an important role in the origin and maintenance of functional genetic changes within promoter regions. This work is currently being written up for publication (Balhoff, Nielsen, and Wray, in prep.). Experimental analyses of Endo16 protein function: We also carried out a functional analysis of the Endo16 protein to aid in interpreting the results of the described above. We knocked down Endo16 protein in Lytechinus variegatus embryos using morpholino antisense oligonucleotides. We found that Endo16 protein is required for gastrulation cell movements, consistent with its domain of expression in the embryo and its extracellular localization. We also showed that knocking out Endo16 protein does not affect the expression of other genes in the early embryo, suggesting that the observed phenotype is a direct, rather than indirect, effect of the knock-down. Together, these experiments demonstrate the crucial role of Endo16 promoter sequences in the functioning of the gene. This work is now in press (Romano and Wray 2006). Additional activities I published three reviews on the evolution of transcriptional regulation which highlight our NASA-funded research (Wray 2003; Wray et al. 2003; Wray 2006). I also co-edited a methods book for experimental approaches to studying development in non-vertebrate deuterostomes (Ettensohn, Wessel and Wray, 2004). In addition, I presented the results of this research at 9 national and international scientific conferences, and in 12 invited talks to at various universities in the U.S., Canada, and Italy.
Bibliography Type:	Description: (Last Updated: 08/31/2006)
Articles in Peer-reviewed Journals	Balhoff JP, Wray GA. "Evolutionary analysis of the well characterized endo16 promoter reveals substantial variation within functional sites." Proc Natl Acad Sci U S A. 2005 Jun 14;102(24):8591-6. Epub 2005 Jun 3. PMID: 15937122 , Jun-2005

Articles in Peer-reviewed Journals	Wray GA. "The evolution of embryonic gene expression in sea urchins." Integr. Comp. Biol., 2006 Jun; 46(3): 233-42. , Jun-2006
Articles in Peer-reviewed Journals	Romano L, Wray GA. "Endo16 is required for gastrulation in the sea urchin Lytechinus variegatus." Dev Growth Differ. in press, 2006. , Sep-2006
Books/Book Chapters	Ettensohn CA, Wray GA, Wessel GM (eds.). "Development of sea urchins, ascidians, and other invertebrate deuterostomes : experimental approaches." Ed. Charles A. Ettensohn, Gregory A. Wray, Gary M. Wessel. Amsterdam ; Boston : Elsevier Academic Press, c2004. Series: Methods in cell biology, v. 74., Dec-2004